Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I expect to look back on the previous assignments mostly about drafter to ensure my code is written in the correct order. This project will use variables, computation, conditionals, lists, loops, dataclasses, functions, and drafter. I want to improve on my debugging as typically I get stuck when looking for a small difference between what I want the code to do and what actually happens.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the easiest part of the assignment to be creating the dataclasses and initial input page as it will mostly be buttons and input fields with little display. I think the hardest part of this project will be coding the graph display and storing the previously given scenarios temporarily(until the application is closed).

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think that the task that will require the most time and effort will be the graph display portion of the website as we haven't done very much surrounding displaying graphs. I will first implement my dataclasses before implementing the storage, then the function that does the calculating, then the function that displays the calculation as a graph.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions that might interfere with my work ability would be the many other final assignments for my classes that also require deep thinking. In the requirements it says 5 pages is the minimum, but does that mean 5 unique pages to scroll between or 5 different page displays that can be called onto the index?

1) Site Diagram

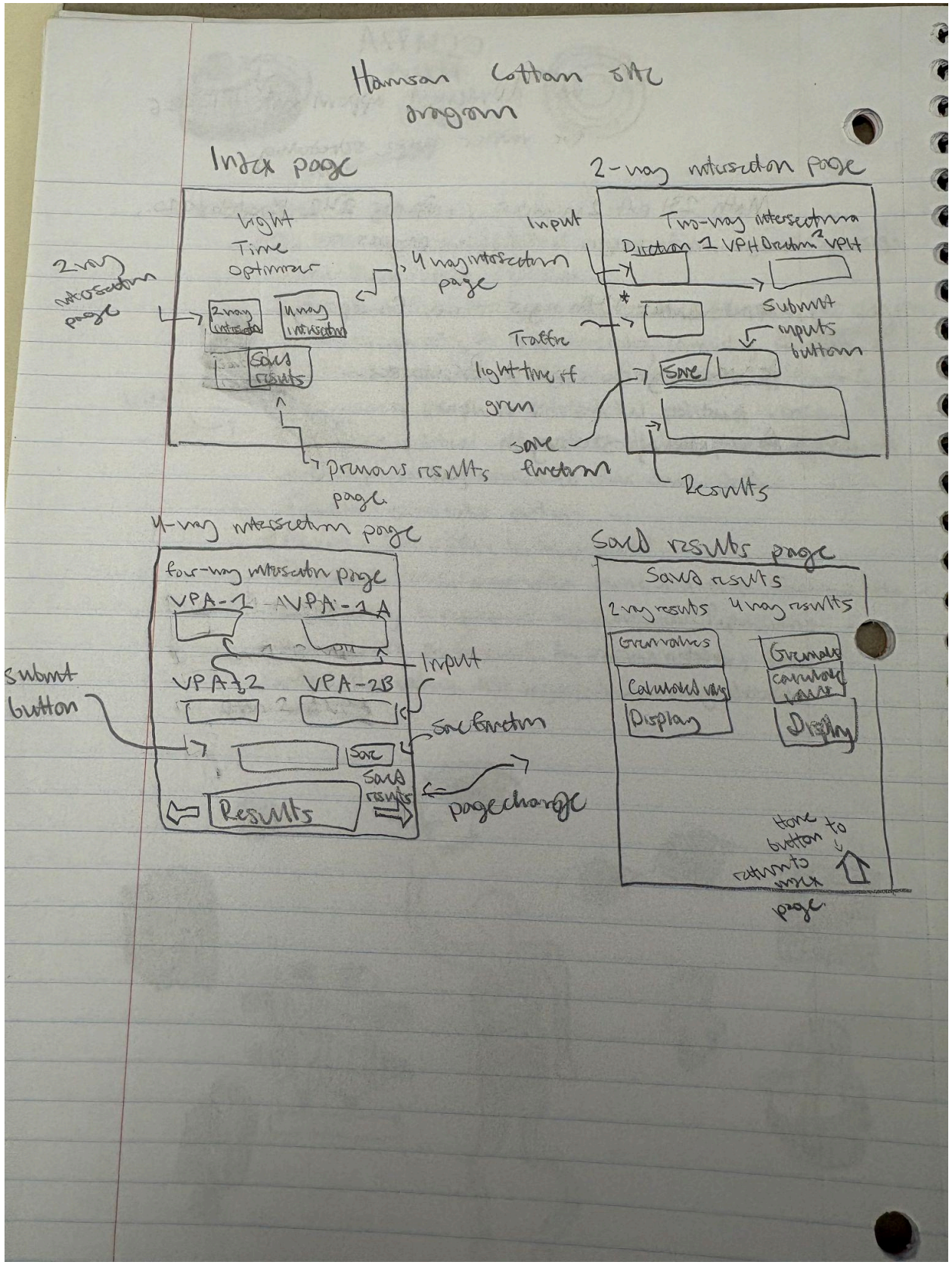
Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.

Hamsan Cottam site diagram



2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**The site will take the direction VPH for one of the two intersection patterns and calculate the theoretical best green light time.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	**Saved Scenarios: List(Scenario)** , used to store previous scenarios in one instance. **Current Intersection name: str**, ensures no intersections get mixed up in storage and easier for the user to see specific scenarios if named. **Selected Intersection Pattern: str**, makes sure the calculations for 2-way and 4-way patterns don't get mixed up. **Current Cycle Length: float**, the signal cycle duration if given from the user.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	**Scenario:** **North_south_volume: int**, vehicles per hour, it will be changed by the users inputs **East_west_volume: int,** vehicles per hour, it would be changed by user inputs. **Calculated percentages: float**, the share of the traffic volume for each lane. **Suggested timings: float,** the estimated green light duration for least traffic.
Routes (minimum 7)	**Home-** brings the user back to the home page from any of the other pages.

The names of your route functions	**Intersection_info-** collects the intersection name and pattern **traffic_volumes-**collects vehicles per hour in each direction **Cycle_length-**collects the total signal cycle length **Calculate_timings:** processes the traffic information and calculates suggested timings **Save_scenario:** saves the calculation as a scenario **saved _scenerios:** displays the previously saved scenarios **Compare_scenarios:** lets the user compare saved scenarios **About_estimes:** explains assumptions and limitations of estimates.
Pages (minimum 5) A brief description of the pages in your site	**Home page(home)-** displays the purpose of the site and explains what information needs to be inputted for calculation. **Intersection Information Page(intersection_info)-**textbox for intersection name, selectbox for intersection pattern, button to continue. **Traffic information page(traffic_input):** textboxes for vehicles per hour in each direction, textbox for total signal cycle length, button to calculate. **Results page(results):** displays traffic volumes and their percentage of total traffic, displays suggested green light timings, button to save the scenario.\ **Saved Scenarios Page(saved_scenarios):** displays previously saved intersections, allows users to compare scenarios, displays a small visual representing the scenario.
User Input (minimum 3) Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes	**Intersection name, textbox, name of the intersection** **Intersection pattern: selectbox, type of intersection** **North/South traffic volume: textbox, vehicles per hour**

For more information on these input types, see Drafter Documentation.	**East/West traffic volume: textbox, vehicles per hour** **Signal cycle length: Textbox, total cycle length in seconds**
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	One will validate traffic volume by ensuring that it is not negative, it will validate cycle length greater than zero, and it will check saved scenarios to see if there are any and display a message if the page is called but there are no saved results.
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	The loop will iterate through saved scenario dataclasses and display their information for comparison.

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

https://chatgpt.com/share/6aecf78b-fb8c-83ea-82d3-967d09fca14b

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.